

HOVASC TABLET

VIHOV13-0var (MY)_2

DESCRIPTION

Hovasc 5mg tablet

Oblong, white to off white uncoated tablet; bevel edged, shallow convex with break bar on one face and "HOVID" embossed on another face.

Hovasc 10mg tablet

Oblong, white to off white uncoated tablet; bevel edged, shallow convex with break bar on one face and "HOVID" embossed on the other face.

COMPOSITION

Each uncoated tablet contains:

Hovasc 5mg tablet

Amlodipine Besilate equivalent to Amlodipine 5mg

Hovasc 10mg tablet

Amlodipine Besilate equivalent to Amlodipine 10mg

ACTIONS AND PHARMACOLOGY

Amlodipine is a dihydropyridine calcium antagonist. It acts by selective inhibition of transmembrane influx of calcium ions into the cardiac muscle and vascular smooth muscle through specific ion channels; thus modulating the contractile process of these cells. Amlodipine has a greater effect on vascular smooth muscle cells than on cardiac muscle cells.

Amlodipine is a peripheral arterial vasodilator that acts directly on vascular smooth muscle to cause a reduction in peripheral vascular resistance and reduction in blood pressure.

Amlodipine exhibits negative inotropic effects *in vivo*, but appears to have no significant effect on the sinoatrial or atrioventricular node in humans.

Although not fully defined, the mechanism of action of amlodipine in relieving angina is thought to be as follows:

Exertional Angina

Amlodipine reduces the total peripheral resistance (afterload) against which the heart works and reduces the rate pressure product. Therefore, in patients with exertional angina, myocardial oxygen demand is reduced, at any given level of exercise.

Vasospastic Angina

Amlodipine has been demonstrated to block constriction and restore blood flow in coronary arteries and arterioles in response to calcium, potassium epinephrine, serotonin, and thromboxane A₂ analog in experimental animal models and in human coronary vessels *in vitro*. This inhibition of coronary spasm is responsible for the effectiveness of amlodipine in vasospastic (Prinzmetal's or variant) angina.

PHARMACOKINETICS

Absorption

Amlodipine is absorbed slowly and almost completely from the gastrointestinal tract. The absorption of amlodipine is not altered by the presence of food. Bioavailability has been estimated to be between 64 and 90%.

Distribution

Plasma concentrations peak between 6 and 12 hours post-dose. *Ex vivo* studies have shown that approximately 93% of the circulating drug is bound to plasma proteins in hypertensive patients. Elimination from the plasma is biphasic with a terminal elimination half-life of about 30-50 hours. Steady-state plasma levels of amlodipine are reached after 7 to 8 days of consecutive daily dosing.

Metabolism

About 90% of amlodipine is converted into inactive metabolites via hepatic metabolism.

Elimination

10% of the parent compound and 60% of the metabolites are excreted in the urine.

Renal impairment

The pharmacokinetics of amlodipine is not significantly altered by renal impairment. Patients with renal failure may therefore receive the usual initial dose.

Elderly patients and patients with hepatic insufficiency may have decreased clearance of amlodipine and resulting increase in AUC of approximately 40-60%; a lower initial dose may be required. A similar increase in AUC was observed in patients with moderate to severe heart failure.

INDICATIONS

Amlodipine is indicated for the first line treatment of hypertension and can be used as the sole agent to control blood pressure in the majority of patients. Patients not adequately controlled on a single antihypertensive agent may benefit from the addition of amlodipine, which has been used in combination with a thiazide diuretic, alpha blockers, beta adrenoceptor blocking agent, or an angiotensin-converting enzyme inhibitor. Amlodipine is indicated for the first line treatment of myocardial ischemia, whether due to fixed obstruction (stable angina) and/or vasospasm/vasoconstriction (Prinzmetal's or variant angina) of coronary vasculature. Amlodipine may be used where the clinical presentation suggests a possible vasospastic/vasoconstrictive component but where vasospasm/vasoconstriction has not been confirmed. Amlodipine may be used alone, as monotherapy, or in combination with other antianginal drugs in patients with angina that is refractory to nitrates and/or adequate doses of beta blockers.

CONTRAINDICATIONS

Amlodipine is contraindicated in patients with known hypersensitivity to dihydropyridine derivatives.

WARNINGS AND PRECAUTIONS

General

Vasodilation induced by amlodipine is gradual in onset. Even though, acute hypotension has rarely been reported after oral administration of amlodipine, caution should be exercised when administering amlodipine particularly in patients with severe aortic stenosis. In general, calcium channel blockers should be used with caution in patients with heart failure.

Beta-Blocker Withdrawal

Any withdrawal of beta-blocker should be by gradual reduction of the dose. Amlodipine is not a beta-blocker and thus gives no protection against the dangers of abrupt beta-blocker withdrawal;

Patients with Hepatic Failure

Amlodipine should be administered with caution to patients with severe hepatic impairment. Amlodipine is extensively metabolized by the liver and the plasma elimination half-life (t_{1/2}) is 56 hours in patients with impaired hepatic function. Amlodipine was associated with increased reports of pulmonary edema despite no significant difference in the incidence of worsening heart failure as compared to placebo.

Increased Angina and/or Myocardial Infarction

Rarely, increased frequency, duration and/or severity of angina or acute myocardial infarction have been documented in patients, particularly those with severe obstructive coronary artery disease, when started on calcium channel blocker therapy or at the time of dosage increase. The mechanism of this effect has not been defined.

MAIN SIDE/ADVERSE EFFECTS

Amlodipine is well tolerated.

The most commonly observed side effects were:

- **Autonomic Nervous System**
 - flushing
- **Body As A Whole**
 - fatigue
- **Cardiovascular, General**
 - edema
- **Central & Peripheral Nervous System**
 - dizziness, headache
- **Gastrointestinal**
 - abdominal pain, nausea
- **Heart Rate/Rhythm**
 - palpitations
- **Psychiatric**
 - somnolence

Less commonly observed side effects in marketing experience include:

- **Autonomic Nervous**
 - dry mouth, increased sweating
- **Body As A Whole**
 - asthenia, back pain, malaise, pain, weight increase/decrease
- **Cardiovascular, General**
 - hypotension, syncope
- **Central & Peripheral Nervous**
 - hypertonia, hypoesthesia/ paresthesia, peripheral neuropathy, tremor
- **Endocrine**
 - gynecomastia

- **Gastrointestinal**
 - altered bowel habits, dyspepsia (including gastritis), gingival hyperplasia, pancreatitis, vomiting
- **Metabolic / Nutritional**
 - hyperglycemia
- **Musculoskeletal**
 - arthralgia, muscle cramps, myalgia
- **Platelet / Bleeding / Clotting**
 - purpura, thrombocytopenia
- **Psychiatric**
 - impotence, insomnia, mood changes
- **Respiratory**
 - coughing, dyspnea, rhinitis
- **Skin/Appendages**
 - alopecia, skin discoloration, urticaria
- **Special senses**
 - taste perversion, tinnitus
- **Urinary**
 - increased urinary frequency, micturition disorder, nocturia
- **Vascular (Extracardiac)**
 - vasculitis
- **Vision**
 - visual disturbances
- **White Blood Cell / R.E.S.**
 - leucopenia

Rarely, allergic reaction including pruritus, rash, angioedema, and erythema multiforme.

Hepatitis, jaundice and hepatic enzyme elevations have also been reported very infrequently (mostly consistent with cholestasis). Some cases severe enough to require hospitalization have been reported in association with use of amlodipine. In many instances, causal association is uncertain.

As with other calcium channel blockers the following adverse events have been rarely reported and cannot be distinguished from the natural history of the underlying disease: myocardial infarction, arrhythmia (including ventricular tachycardia and atrial fibrillation) and chest pain.

PREGNANCY AND LACTATION

Pregnancy
Studies have not been done in humans. No evidence of teratogenicity or other embryo/fetal toxicity was found in animal studies. Nevertheless, amlodipine should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Nursing Mothers
It is not known whether amlodipine is excreted in human milk. In the absence of this information, it is recommended that nursing be discontinued while amlodipine is administered.

DRUG INTERACTIONS

In vitro data in human plasma indicate that amlodipine has no effect on the protein binding of drugs tested (digoxin, phenytoin, warfarin, and indomethacin).

Cimetidine
Co-administration of amlodipine with cimetidine did not alter the pharmacokinetics of amlodipine.

Sildenafil
A single 100 mg dose of sildenafil in subjects with essential hypertension had no effect on the pharmacokinetic parameters of amlodipine. When amlodipine and sildenafil were used in combination, each agent independently exerted its own blood pressure lowering effect.

Atorvastatin
Co-administration of multiple 10 mg doses of amlodipine with 80 mg of atorvastatin resulted in no significant change in the steady state pharmacokinetic parameters of atorvastatin.

Digoxin
Co-administration of amlodipine with digoxin did not change serum digoxin levels or digoxin renal clearance in normal volunteers.

Ethanol (alcohol)
Single and multiple 10 mg doses of amlodipine had no significant effect on the pharmacokinetics of ethanol.

Warfarin
Co-administration of amlodipine with warfarin did not change the warfarin prothrombin response time.

OVERDOSE

Available data suggest that gross overdosage could result in excessive peripheral vasodilatation and possibly reflex tachycardia. Marked and probably prolonged systemic hypotension up to and including shock with fatal outcome have been reported.

Administration of activated charcoal to healthy volunteers immediately after or up to two hours of amlodipine 10 mg ingestion has been shown to significantly decrease amlodipine absorption. Gastric lavage may be worthwhile in some cases.

Significant hypotension due to overdosage calls for active cardiovascular support including frequent monitoring of cardiac and respiratory function, elevation of extremes, and attention to circulating fluid volume and urine output. A vasoconstrictor may be helpful in restoring vascular tone and blood pressure, provided that there is no contraindication to its use. Intravenous calcium gluconate may be beneficial in reversing the effects of calcium channel blockade. Since amlodipine is highly protein-bound, dialysis is not likely to be of benefit.

DOSAGE AND ADMINISTRATION

For both hypertension and angina the usual initial dose is 5mg amlodipine once daily which may be increased to a maximum dose of 10mg depending on the individual patient's response.

No dose adjustment of amlodipine is required upon concomitant administration of thiazide diuretics, beta-blockers, and angiotensin-converting enzyme inhibitors.

Use in the Elderly
A normal dosage regimen is recommended. Amlodipine, used at similar doses in elderly or younger patients, is equally well tolerated.

Use in Children
Safety and effectiveness of amlodipine in children have not been established.

Use in Patients with Impaired Hepatic Function
As with all calcium antagonists, amlodipine half-life is prolonged in patients with impaired liver function and dosage recommendations have not been established. The drug should therefore be administered with caution in these patients.

Use in Renal Failure
Amlodipine may be used in such patients at normal doses. Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment. Amlodipine is not dialyzable.

Note: The information given here is limited. For further information, kindly consult your doctor or pharmacist.

Storage:
Store below 30°C

Presentation / Packing:
Blisters pack of 3 x 10's, 10 x 10's.

Product Registration Holder: HOVID Bhd.
121, Jalan Tunku Abdul Rahman
(Jalan Kuala Kangsar),
30010 Ipoh, Perak, Malaysia.

Manufactured by: HOVID Bhd.
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31200 Chemor, Perak, Malaysia.

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